

facilitiesHelp.io

Evidence-Graded Trust, Medical-Desert Detection, and Causal Honesty
for 10,000 Messy Healthcare-Facility Records

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Abstract

A planner handed 10,088 scraped Indian healthcare-facility records faces a trust problem: capability claims are unverified, key fields are sparse, and some values are impossible. We present **facilitiesHelp.io**, a Databricks App that treats every field as a *claim to verify, not a fact*. We (i) grade every specialty a facility lists (2,580 distinct; 117,993 graded claims) on evidence drawn from the facility’s own text, attaching a calibrated 0–100 confidence; (ii) validate that the grade reflects *cited evidence rather than facility prominence* (a model predicting the grade from facility metadata alone achieves $AUC \approx 0.57$, near chance); (iii) classify district-level care gaps for 2,518 capabilities, separating a *real* gap from a *data-poor* region; and (iv) build a causal layer on 706 NFHS-5 districts that distinguishes real levers from coincidences. Our headline causal result: the sanitation–stunting association ($r = -0.51$) *collapses to ≈ 0 after adjusting for household wealth*, while female-schooling→child-marriage ($r = -0.65$) and antenatal-care→institutional-birth ($r = +0.60$) survive. Crucially, facility *count* is only weakly linked to outcomes after adjustment, which reframes the supply–demand question for planners. Every claim cites its source text; uncertainty is communicated, never hidden.

1 Introduction

The challenge dataset contains 10,088 Indian healthcare facilities described by 51 columns: structured fields (name, location, controlled specialties) plus *uneven free-text* descriptions of claimed capabilities, procedures, equipment, and services. Field coverage is highly non-uniform (Table 1), the free text is noisy and repetitive, and some structured values are impossible. The user—a non-technical planner, NGO coordinator, or analyst—must nonetheless make decisions they can trust: *Can this facility do what it claims? Where are the real care gaps? Where should a patient go? What must be fixed before the data is trustworthy?*

Our contributions:

1. A deterministic, auditable **trust-grading engine** over the facility’s own evidence, with a calibrated confidence and a contradiction check (§3).
2. A **validation** that the grade is evidence-driven, not a popularity proxy (§4).
3. A **medical-desert** classifier that separates real gaps from data-poor regions for 2,518 capabilities, addressing the supply–demand challenge (§5).

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4. A **correlation–causation** layer on 706 NFHS districts using structure learning, multi-estimator adjustment, and sensitivity analysis (§6–7), with statistical-ML and geometric-DL extensions (§8).

2 Datasets and granular profiling

We use three read-only Unity Catalog datasets:

- **facilities** — 10,088 rows $\times$ 51 columns; the core, treated as claims to verify.
- **india_post_pincode_directory** — 165,627 post offices with district, state, latitude, longitude; powers geocoding and the facility→district join.
- **nfhs_5_district_health_indicators** — 706 districts $\times$ 109 indicators; powers the causal layer.

Coverage and missingness. Substantive (non-empty content) coverage of the evidence fields is uneven (Table 1); 9,996/10,088 records carry a postcode. We treat missingness honestly: sparse fields are shown as “*not reported*,” never imputed, and the confidence of a claim reflects how much evidence actually exists.

Field	Coverage	Field	Coverage
description	100%	numberDoctors	36.4%
capability	99.7%	capacity (beds)	25.2%
procedure	92.5%	yearEstablished	47.8%
equipment	77.0%	postcode	99.0%

Table 1: Evidence-field coverage. Useful signal is spread across *description*, *capability*, *procedure*, *equipment*, *specialties*, and *source_urls*; all are claims, not ground truth.

Entity resolution. The same facility is scraped from many sources. We deduplicate to one canonical record per pre-computed match key `cluster_id` via `row_number()` OVER (PARTITION BY `cluster_id` ORDER BY `source_urls`) collapsing 10,088 raw rows to 9,959 facilities. A medallion pipeline (bronze→silver→gold) on a serverless SQL Warehouse then assembles the evidence text, counts independent source types, and joins facilities→PIN→district.

3 The trust-grading engine

For each (facility, capability) we test two evidence signals against the facility’s own text and count independent sources:

$$\text{spec_hit} = \mathbb{1}[\text{spec_text} \sim r_{\text{spec}}], \quad \text{text_hit} = \mathbb{1}[\text{evidence_text} \sim r_{\text{kw}}], \quad n_{\text{src}} = |\text{distinct source_types}|,$$

where `evidence_text` = `lower(description || capability || procedure || equipment)` and $\sim$ denotes a regex match. The grade is

$$\text{grade} = \begin{cases} \text{WEAK/SUSPICIOUS} & \text{facility type contradicts an acute claim} \\ \text{STRONG} & \text{spec.hit} \wedge \text{text.hit} \\ \text{PARTIAL} & \text{spec.hit} \vee \text{text.hit} \\ \text{NO CLAIM} & \text{otherwise (dropped)} \end{cases}$$

The contradiction rule flags a small `clinic/dentist/doctor/pharmacy` asserting an acute capability (ICU, oncology, ...) in free text with *no* matching specialty code. Confidence is a bounded, monotone function of corroboration:

$$\text{conf} = \begin{cases} \min(95, 70 + 5 n_{\text{src}}) & \text{STRONG} \\ 50 & \text{PARTIAL} \\ 20 & \text{WEAK/SUSPICIOUS} \end{cases}$$

We apply this enhanced rubric (with the contradiction check) to 22 high-stakes acute capabilities, and a generalized version to *every* specialty a facility lists, yielding 117,993 graded specialty-claims across 2,580 specialties (Table 2). Every grade is accompanied by the verbatim evidence and ranked source links, so a planner can audit it.

Grade	22 acute capabilities	All 2,580 specialties
STRONG	36,847	40,151
PARTIAL	37,318	64,898
WEAK/SUSPICIOUS	1,170	—
CLAIMED (listed only)	—	12,944
Total	75,335	117,993

Table 2: Graded-claim distribution. STRONG requires ≥ 2 independent evidence types and corroboration; most claims are PARTIAL, reflecting honest single-signal evidence.

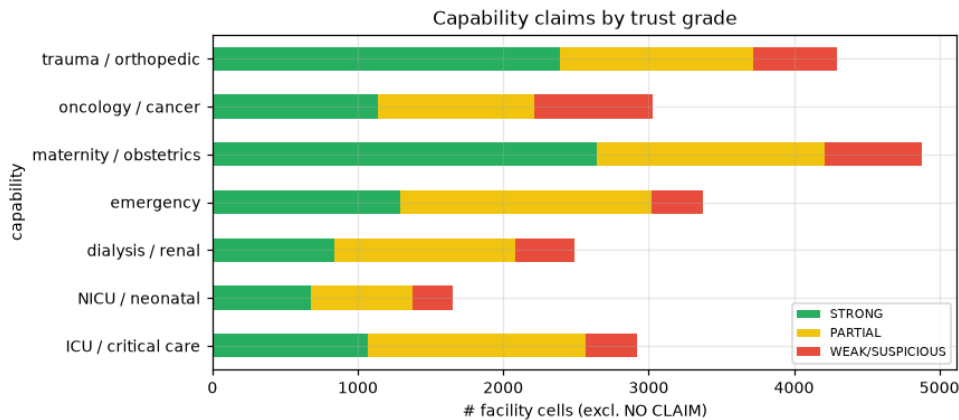


Figure 1: Evidence grades by capability. Maternity is best-evidenced; ICU/NICU are rarest—visibility, not necessarily absence.

4 Validating the grade: reproducibility and non-popularity

Two questions: is the grade a faithful function of its evidence, and is it merely a proxy for facility prominence?

1. **Reproducibility.** A logistic regression $\Pr(\text{STRONG}) \sim \text{spec_hit} + \text{text_hit} + n_{\text{src}}$ attains 5-fold ROC-AUC = 1.00, confirming the grade is a transparent, deterministic function of its evidence (attribution $\approx 49\%$ specialty code, $\approx 51\%$ procedure/equipment text).
2. **Non-popularity.** A gradient-boosting classifier predicting **STRONG** from facility *metadata only* (capacity, doctors, web presence, #specialties, facility type)—*excluding* the rule inputs—scores only 4-fold ROC-AUC = 0.57, barely above chance. Therefore the grade reflects *cited evidence, not size or fame*.

We additionally expose a SHAP-style per-claim attribution (Fig. 2) that decomposes each confidence score into its contributing signals.

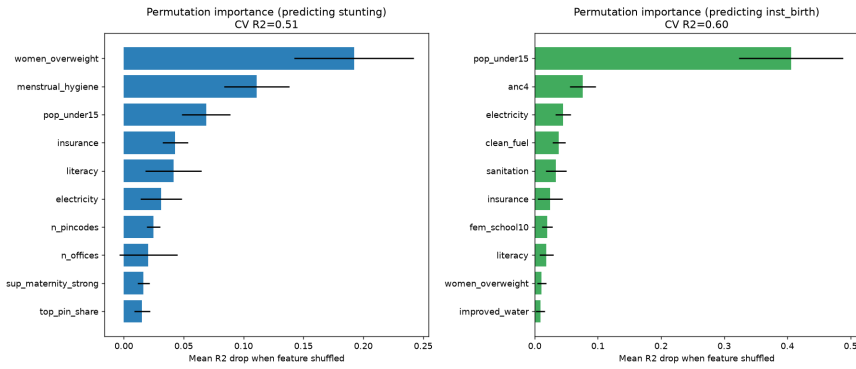


Figure 2: Feature importance / attribution for the evidence model.

5 Medical deserts and the supply–demand challenge

For each (capability, district) aggregated over distinct facilities we classify

$$\text{class} = \begin{cases} \text{DATA-POOR} & |\{\text{facilities}\}| < 3 \\ \text{APPARENT CARE GAP} & n_{\text{STRONG}} < 2 \\ \text{EVIDENCED SUPPLY} & \text{otherwise.} \end{cases}$$

This is the honesty hinge: it separates a *real* gap (facilities exist but little trustworthy evidence) from a region we simply *under-measured* (Table 3). Pre-computed for all 2,518 capabilities. That the modal cell is DATA-POOR is itself the finding—most district×capability combinations cannot be judged from scraped data, and we say so rather than declaring “no care.”

District×capability classification	Count
DATA-POOR (can’t tell)	24,824
EVIDENCED SUPPLY	8,674
APPARENT CARE GAP (real)	4,675

Table 3: Care-gap classification across 2,518 capabilities and 706 districts.

Supply vs. demand. Facility data is *supply* (what care exists, how trustworthy); NFHS is *demand and outcomes*. They meet at the district key. The pivotal question—*will adding facilities improve outcomes?*—is answered in §7: facility *count* is only weakly linked to outcomes after adjustment, so a desert is necessary but not sufficient for an investment decision.

6 Correlation analysis

We summarize linear association with the Pearson coefficient

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} \in [-1, 1],$$

computed across the 706 districts of `gold_nfhs`; each row is a district and each column an NFHS percentage. Strength tiers: $|r| < 0.1$ none, 0.1–0.3 weak, 0.3–0.5 moderate, 0.5–0.7 strong, > 0.7 very strong. Selected live values appear in Table 4; the full matrix is the interactive heatmap (Fig. 3).

Pair (district level, $n = 706$)	r
sanitation $\leftrightarrow$ stunting	−0.51
clean fuel $\leftrightarrow$ stunting	−0.42
female schooling $\leftrightarrow$ child marriage	−0.65
ANC4 visits $\leftrightarrow$ institutional birth	+0.60
women overweight $\leftrightarrow$ high blood sugar	+0.62
wealth $\leftrightarrow$ sanitation	+0.32
wealth $\leftrightarrow$ stunting	−0.43

Table 4: Pearson correlations. The last two rows are the “smoking gun”: wealth is positively linked to sanitation *and* negatively to stunting, which mechanically manufactures the sanitation–stunting association.

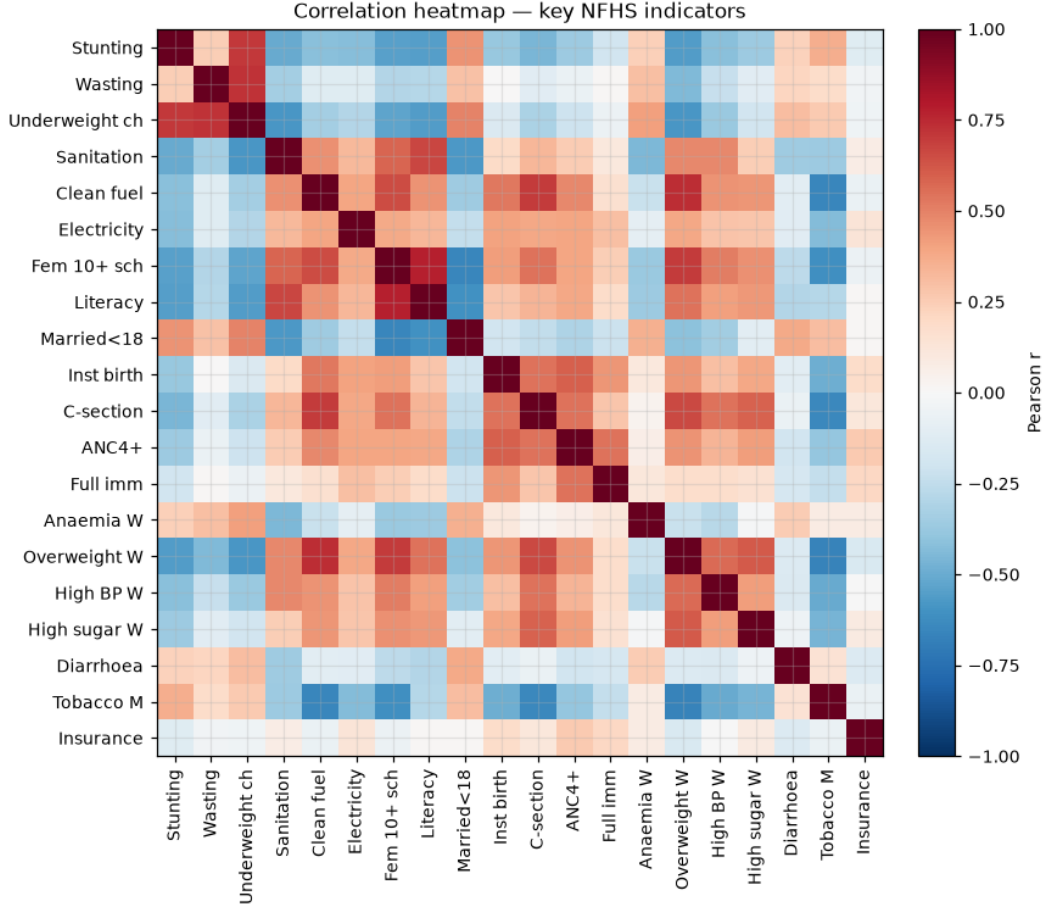


Figure 3: NFHS-5 correlation matrix across 706 districts (red positive, blue negative).

7 Causal inference: separating levers from coincidences

Confounding. A confounder Z causes both X and Y , fabricating an X – Y association even when X does not cause Y . Household **wealth** is the canonical example here: wealthier districts install more toilets ($\uparrow$ sanitation) *and* feed children better ($\downarrow$ stunting), so sanitation and stunting correlate without a direct causal link. Table 4 confirms the mechanism quantitatively.

The causal ladder. We climb from association to a defensible effect:

1. **Structure learning** — the PC algorithm (constraint-based conditional-independence tests) infers a DAG (Fig. 4), identifying which variables to adjust for.
2. **OLS with state fixed effects** — $Y = \beta X + \gamma^\top Z + \sum_s \delta_s \mathbb{I}[\text{state} = s] + \varepsilon$; read $\hat{\beta}$ holding wealth and state constant.
3. **Double Machine Learning** — partial out confounders from both treatment and outcome with ML and regress the residuals (Frisch–Waugh–Lovell) with cross-fitting, robust to nonlinear confounding.

4. **Propensity-score matching** — match high- vs. low- X districts on $\hat{e}(Z) = \Pr(X=1 | Z)$ and compare outcomes.
5. **Sensitivity (E-value)** — the minimum strength an unmeasured confounder would need with both X and Y to nullify the effect; large = robust.

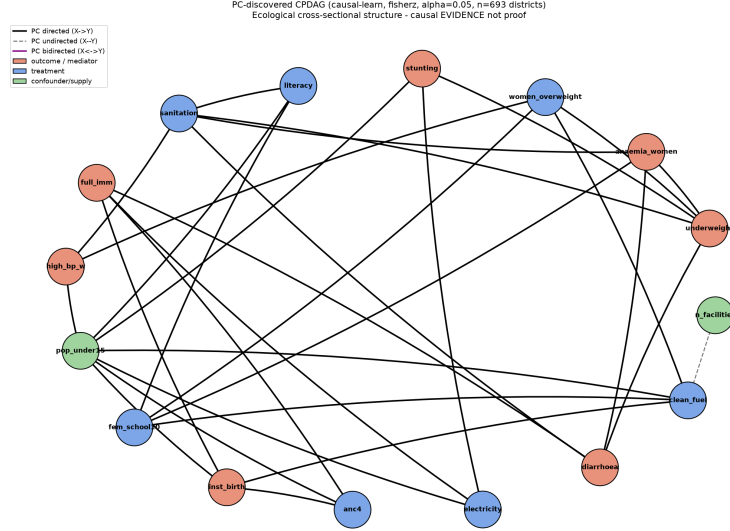


Figure 4: PC-learned causal DAG over NFHS indicators (treatments, mediators, confounders, outcomes).

Results. Standardized effects across estimators are summarized in Table 5 and Figs. 5–6. The decisive pattern: effects that *collapse toward zero* under adjustment were confounded; effects that *survive* are likely causal.

Relationship	Naive r	OLS+FE	Double-ML	PSM	Verdict
sanitation $\rightarrow$ stunting	-0.51	-0.15	-0.11	-0.01	confounded
clean fuel $\rightarrow$ stunting	-0.42	-0.16	-0.14	-0.06	mostly confounded
schooling $\rightarrow$ child marriage	-0.65	-0.18	-0.17	-0.40	likely causal
ANC4 $\rightarrow$ institutional birth	+0.60	+0.16	+0.13	+0.36	likely causal
overweight $\rightarrow$ high blood pressure	+0.57	+0.29	+0.35	+0.44	causal

Table 5: Naive vs. adjusted standardized effects. Sanitation $\rightarrow$ stunting vanishes; schooling, ANC4, and overweight effects persist.

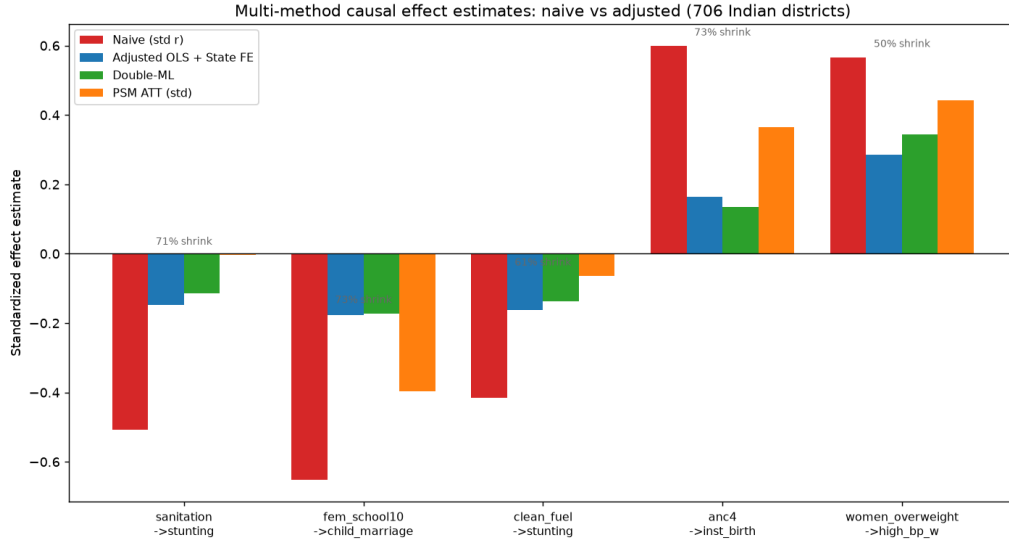


Figure 5: Correlation $\rightarrow$ causation across four estimators. Bars collapsing to 0 were confounded; bars that hold are likely causal.

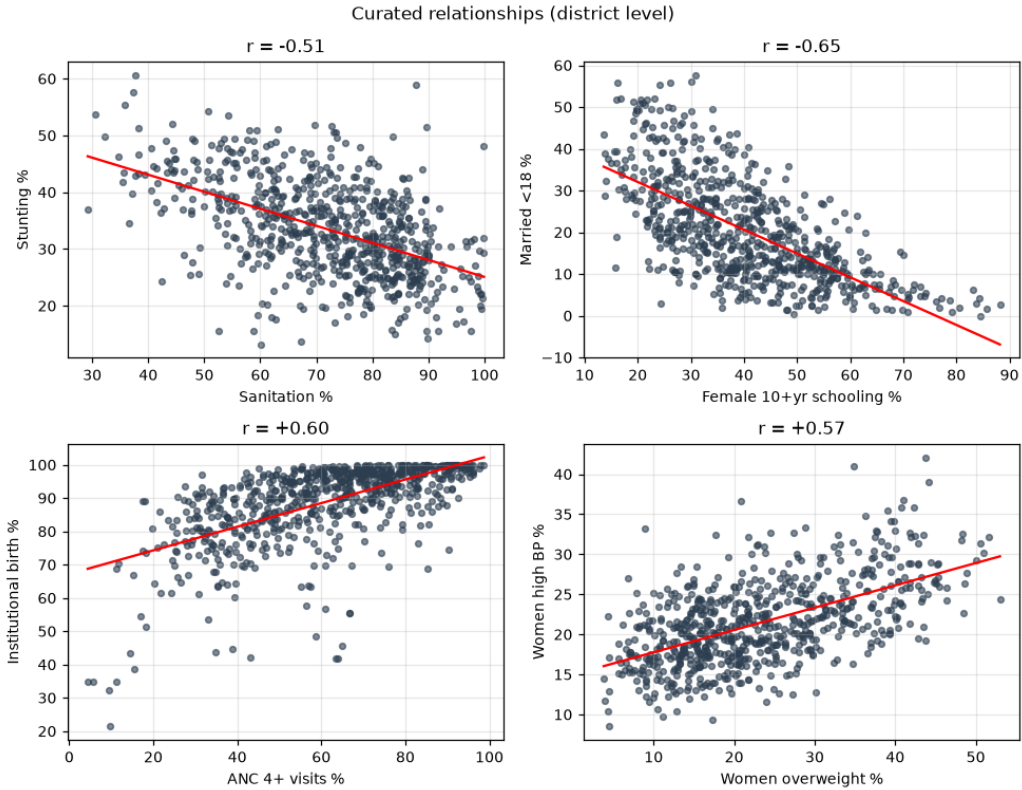


Figure 6: Naive vs. wealth-adjusted scatters; the apparent sanitation–stunting slope flattens once wealth is accounted for.

The result that connects to facilities. The same machinery shows facility *count* is only weakly linked to outcomes after adjustment. Hence “build more facilities” is not automatically the fix; the

binding lever is often demand-side (schooling, antenatal care). This is precisely the honesty a “for good” planning tool owes its user.

Interactive exploration. The DAGs are explorable in-app through a force-directed causal-graph studio, with `d3.js` *inlined* into the Databricks App to satisfy its content-security policy (a CDN-loaded library is blocked). Six modules—the full NFHS map, the wealth-confounding fork, the trust evidence→verdict graph, the care-pathway co-occurrence graph, supply-vs-demand, and the maternal/adolescent chain—offer draggable nodes, zoom, and per-edge evidence on hover.

8 Statistical ML and geometric deep learning

Beyond linear adjustment we ran: **GAM + quantile regression** for nonlinear dose–response and high-burden-district (tail) effects (Fig. 7); **multilevel/hierarchical** models with empirical-Bayes shrinkage to stabilize small-sample districts; penalized regression with **stability selection**; and **geometric deep learning**—a spatial GNN with positional encoding and a Tobler’s-Law smoothing penalty—benchmarked against baselines (Fig. 8). Moran’s I is used as a spatial-autocorrelation *diagnostic*, not as proof of a spatial process.

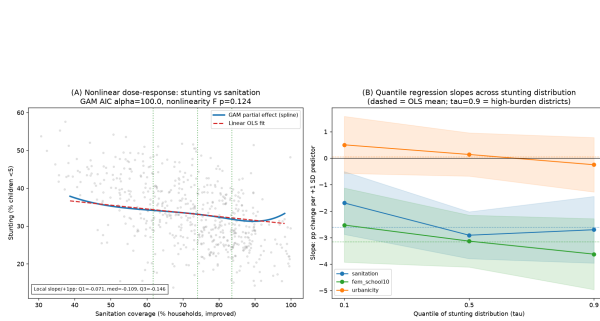


Figure 7: GAM + quantile regression.

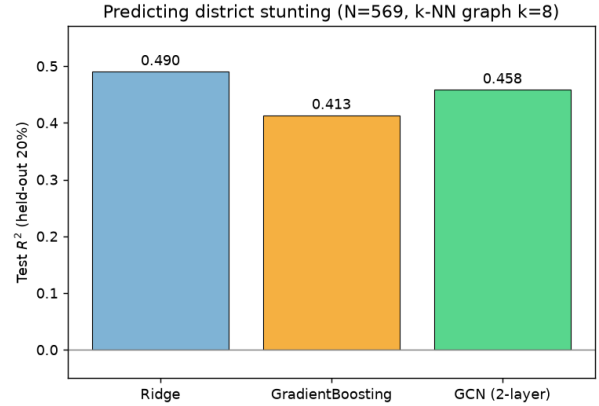


Figure 8: Spatial GNN vs. baselines.

9 Geospatial methods and referral

We geocode *any* city/district to a centroid from the 165,627-office India-Post directory (coordinates bounded to India, 6–38°N × 67–98°E, to reject corrupt values) and rank facilities evidence-first, then by great-circle distance

$$d = 6371 \cdot \arccos(\sin \phi_1 \sin \phi_2 + \cos \phi_1 \cos \phi_2 \cos(\lambda_2 - \lambda_1)) \text{ km.}$$

Specialty co-occurrence (“you may also need”). Beyond the requested specialty A , we rank related specialties B by the empirical conditional probability that a facility offering A also offers B ,

$$P(B | A) = \frac{\#\{\text{facilities offering both } A \text{ and } B\}}{\#\{\text{facilities offering } A\}},$$

materialized in `gold_specialty_cooccur` over all facilities (kept where $\#A \geq 20$, $\#\{A, B\} \geq 10$ and $P(B | A) \geq 0.15$; 6,214 ordered pairs). For instance $P(\text{Internal Medicine} | \text{Cardiology}) = 0.93$ and $P(\text{Orthopedic Surgery} | \text{Cardiology}) = 0.73$. This is a supply-side *bundling* signal, surfaced as a soft suggestion, not medical advice.

Care pathways: cause $\rightarrow$ related needs. A patient routed to a specialty *because of* a condition often needs care for *co-occurring* conditions. We ground this in district-level correlations among the seven patient-facing NFHS indicators (`gold_condition_corr`, all pairs, $n \approx 705$). For a chosen concern we report sibling conditions within the same clinical cluster — child nutrition (stunting, wasting, child anaemia), metabolic (overweight, high blood sugar), maternal (ANC4, institutional birth) — annotated with the measured Pearson r . Child stunting travels with child anaemia ($r = +0.33$) and wasting ($r = +0.25$); a stunting referral to a paediatrician/child-nutritionist therefore also flags anaemia screening (paediatrics + pathology) and nutrition care nearby, each labelled a *maybe* with an explicit **population correlation $\neq$ individual diagnosis** caveat. This is where the causal layer (§7) meets the referral layer: the correlations are real and shown, the confounders (e.g. wealth) are examined separately, and the clinical action is *offered, never asserted*.

Roadmap: 2SFCA accessibility scoring to weight reachable supply within a travel radius; and *last-mile* integrations that turn a verdict into an action—open or route to a facility in Google Maps, place a call or WhatsApp message to it, and email a shortlist or referral letter to a colleague—so a trusted recommendation becomes a next step in one tap.

10 Honesty, uncertainty, and limitations

Every claim cites the facility’s own text; confidence is always shown; the assistant refuses to invent numbers, answering only from grounded gold-table data. Limitations: NFHS-5 is *cross-sectional* (one wave), so our adjusted associations are the strongest single-wave causal claim—two waves would enable difference-in-differences. Scraped supply data carries visibility bias. Suppressed NFHS values are MNAR and are flagged, never imputed.

11 Conclusion

facilitiesHelp.io turns 10,088 messy records into decisions a non-technical planner can trust, across all four challenge tracks plus two bonus tracks. The technical core is honesty: grade on cited evidence, communicate uncertainty, separate a real care gap from missing data, and distinguish a correlation from a cause. The hardest part of “for good” data work is not prediction—it is telling the user what the data can and cannot support.

A Data dictionary (gold tables)

Table	Purpose
silver_facility	deduped facility; evidence text; PIN→district
gold_facility_capability_trust	22 acute capabilities + contradiction check (75,335)
gold_facility_specialty	every listed specialty graded (2,580; 117,993)
gold_facility_contact	address, phone, web, beds, doctors, equipment
gold_all_gaps, gold*_district_gaps	care-gap classification (2,518 caps)
gold_referral	distance-rankable referral index
gold_specialty_cooccur	specialty co-occurrence $P(B A)$ (6,214 pairs)
gold_condition_corr	NFHS condition correlations for care pathways
gold_nfhs, gold_pin_state, gold_specialty_counts	analysis aggregates
app_user_actions	persisted overrides/shortlists/scenarios/chats

Live app: <https://virtue-desk-7474648370355072.aws.databricksapps.com>

Code: <https://github.com/ckwan48/databricks-hackathon-2026>

Stack: Databricks Free Edition — Unity Catalog, serverless SQL Warehouse, Delta medallion, Databricks Apps (Streamlit), Model Serving (Llama 4 Maverick), AI/BI Genie, Databricks agent skills.